al-nafi INTERNATIONAL COLLEGE

# Exam Topic for Oral Presentation
# EduQual Level 6

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**Topic 59: Building Interplanetary Communication Network with Delay-Tolerant Networking, Quantum Communication, and Space-Based Infrastructure for Mars Mission Support**

1. **Overview:**
   This project requires students to design and implement a comprehensive interplanetary communication network supporting delay-tolerant networking, quantum communication, and space-based infrastructure for Mars mission support and deep space exploration. The solution must handle extreme communication delays, radiation effects, and autonomous operation while ensuring reliable data transmission across interplanetary distances. Students will build a space communication platform combining advanced networking protocols, quantum technologies, and distributed space systems.

2. **Key Learning Objectives:**
   a. **Delay-Tolerant Networking (DTN) Protocols:**
      i. Deploy Bundle Protocol and DTN routing algorithms for space communication
      ii. Implement store-and-forward mechanisms for interplanetary data relay
      iii. Configure adaptive routing based on orbital mechanics and link availability
   b. **Quantum Communication and Cryptography:**
      i. Deploy quantum key distribution (QKD) for secure space communications
      ii. Implement quantum repeaters and entanglement distribution protocols
      iii. Configure post-quantum cryptography for long-term security
   c. **Space-Based Network Infrastructure:**
      i. Deploy satellite constellation design for Mars-Earth communication relay
      ii. Implement ground station networks and deep space network integration
      iii. Configure lunar gateway and Lagrange point communication nodes
   d. **Orbital Mechanics and Link Prediction:**
      i. Deploy orbital propagation models for communication window prediction
      ii. Implement Doppler shift compensation and link budget calculations
      iii. Configure autonomous spacecraft communication and scheduling
   e. **Radiation-Hardened Computing Systems:**
      i. Deploy radiation-tolerant computing platforms for space environments
      ii. Implement error detection and correction for space-based data processing
      iii. Configure autonomous system recovery and fault tolerance mechanisms
   f. **Mission-Critical Data Prioritization:**
      i. Deploy intelligent data routing and prioritization algorithms
      ii. Implement compression and data reduction techniques for bandwidth optimization

iii. Configure emergency communication protocols and backup systems

g. **Industry Application:** Used by space agencies, aerospace companies, satellite communication providers, and deep space exploration missions implementing interplanetary communication and space-based internet infrastructure.

h. **Open Source and Cloud-Native Tools:**
   i. **DTN:** ION-DTN, JXTA, Epidemic routing protocols
   ii. **Quantum:** Qiskit, QuTiP, Strawberry Fields
   iii. **Orbital Mechanics:** GMAT, Orekit, STK
   iv. **Space Systems:** cFS (Core Flight System), COSMOS
   v. **Compression:** CCSDS standards, JPEG 2000, LZMA
   vi. **Simulation:** ns-3, OMNeT++, MATLAB Simulink
   vii. **Standards:** CCSDS, ITU-R space protocols, DVB-S2X

3. **Examination Format:**
   a. **Presentation (15–20 minutes):**
      i. Interplanetary communication network architecture with DTN protocols
      ii. Quantum communication implementation and security analysis
      iii. Space-based infrastructure and orbital mechanics integration
      iv. Mars mission communication scenario and performance evaluation
   b. **Interview (30–40 minutes):**
      i. Space communication challenges and delay-tolerant networking principles
      ii. Quantum communication applications and space-based quantum technologies
      iii. Orbital mechanics and deep space mission communication requirements
      iv. Various questions will be asked as per the topic scope.

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